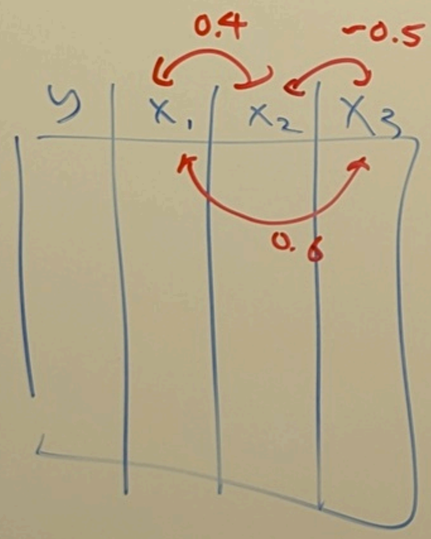
Multicolinearity

$$y = \beta_0 + \beta_1 X_1 + \ldots + \beta_d X_d + \varepsilon$$

corr ≈ 1 ?

predictors



$lm(X_1 \sim X_2 + X_3)$

$lm(X_2 \sim X_1 + X_3)$

$lm(X_3 \sim X_1 + X_2)$

$corr(\underline{1}X_1 + \underline{0.8}X_2, X_3) \approx \underline{1}$

$?X_1 + ?X_2 \approx X_3$

$corr \approx 0.9$

$R^2 = corr^2 = 0.81$

$lm(X_3 \sim X_1 + X_2)$

$corr(X_3, 0.97X_1 + 0.78X_2) \approx \underline{1}$

$R^2 = corr^2 \approx \underline{1}$

$$VIF = \frac{1}{1-R^2}$$

$R^2 \approx 0 \quad VIF \simeq 1$

$R^2 \approx 1 \quad VIF \approx \infty$

model's $y = 1.5 + \underbrace{2.57}_{\uparrow} x_1 + \underbrace{0}_{\uparrow} \cdot x_2$

not interpretable

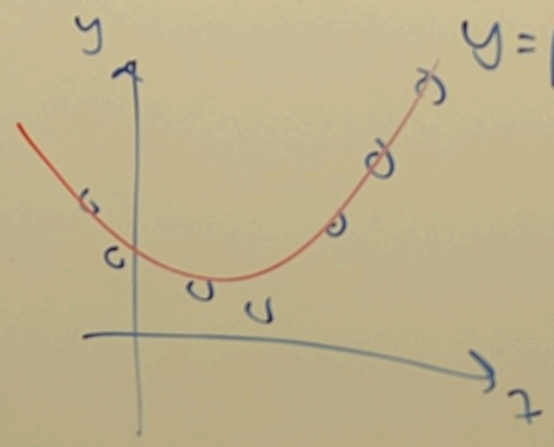
$\hat{y} \leftarrow$ predict() ok

Polynomic Reg

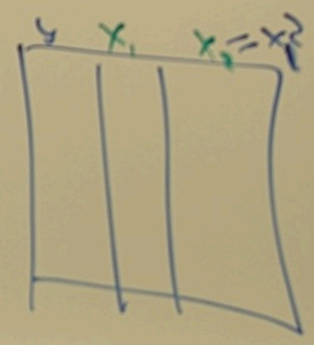
$y = chol$

$x = bp$

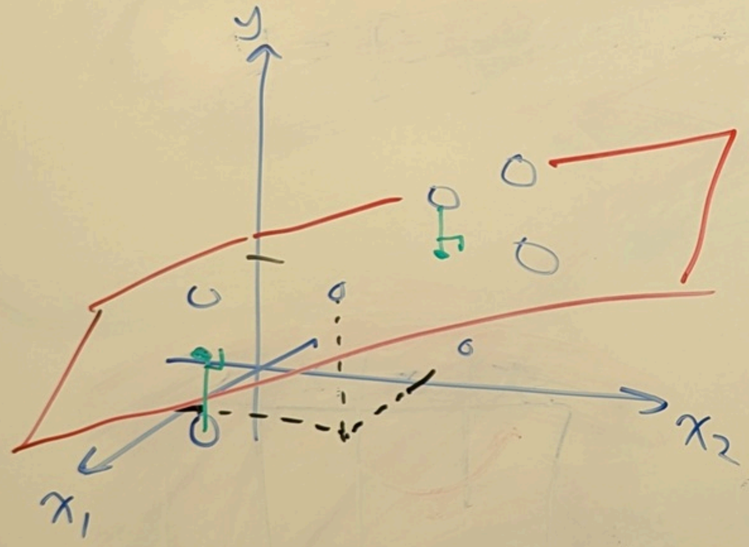
$y = \beta_0 + \beta_1 x + \varepsilon$

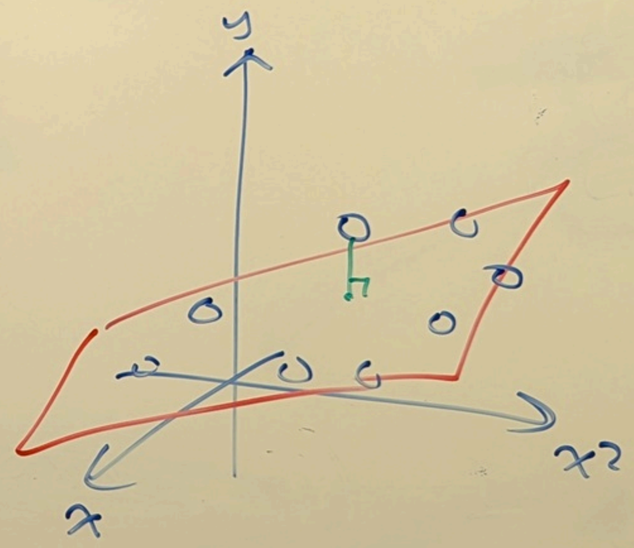


$y = \beta_0 + \beta_1 \underbrace{x}_{=x_1} + \beta_2 \underbrace{x^2}_{=x_2}$

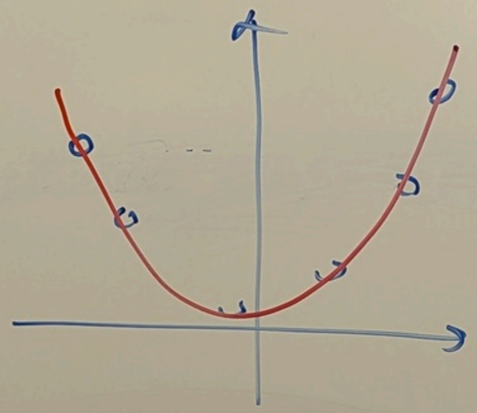


model: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$





$$y = \beta_0 + \beta_1 x + \beta_2 x + \varepsilon$$



$$y = -0.005 + 0.005x + 1.1x^2$$

$\ln(y \sim I(x^2) + 0)$

-1

$x \longmapsto 3 + 2x$

2 levels
sunny / windy

$\ln(\text{feel} \sim \text{temp} + \text{weather})$
?

$\text{sunny} = (1, 0) \in \{0,1\}^{(\#\text{levels})}$

$\text{windy} = (0, 1)$

$\text{rainy} = (0, 0, 1)$

$\ln(\text{feel} \sim \text{temp})$

Indicator

$$x_2 = 1_{\{windy\}} = \begin{cases} 1, & \text{is windy} \\ 0, & \text{not windy} \end{cases}$$

$1_{\{sunny\}}$

model : $y = \underset{11}{\beta_0} + \underset{15.18}{\beta_1}(temp) + \underset{1.0}{\beta_2}(1_{\{windy\}})$ $\quad -30.0$

x_1 $\qquad$ x_2

x_1 = temp

y	temp	weather	x_2
		Sun	0
		Sn	0
		Wind	1

lm(feel ~ temp + weather)

$\vec{y} =$

Back 7:40pm

$$y = \beta_0 + \beta_1 (\text{temp}) + \beta_2 (1_{\{\text{sunny}\}}) + \beta_3 (1_{\{\text{windy}\}})$$

X_1 X_2 X_3

X_0	temp	weather	1_S	1_W
1	⋮	Sun	1	0
		Sun	1	0
		Windy	0	1
1				

$\vec{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

X

$$X_0 = X_2 + X_3$$

$$\text{corr}(X_0, X_2 + X_3) = 1$$

Interactions

$$y = \underset{2.55}{\beta_0} + \underset{1.18}{\beta_1}(\underbrace{temp}_{X_1}) + \underset{-2.89}{\beta_2}(\underbrace{1_{\{windy\}}}_{X_2}) + \underset{-0.38}{\beta_3}(\underbrace{temp \times 1_{\{windy\}}}_{X_3}) + \varepsilon$$

β_3: interaction

ε: random, $N(0, \sigma^2)$

y	X_1 temp	weather	X_2 $1_{\{windy\}}$	X_3
	60	Sun	0	0
	65	Windy	1	65
	⋮	⋮	⋮	⋮

lm(feel ~ temp + weather + temp:weather)

1st order, 1st order, 2nd order

lm(feel ~ temp * weather)

$(1+t)(1+w)$

$1 + t + w + tw$

$$(chol) = \beta_0 + \beta_1(\underbrace{bp}_{x_1}) + \beta_2(\underbrace{bmi}_{x_2}) + \beta_3(\underbrace{bp \times bmi}_{x_3})$$

$$y = 1 + 2x_1 + 3x_2 + 4x_1x_2$$
$$\text{1st} \quad \text{1st} \quad \text{2nd}$$

lm(y ~ x1 * x2)

lm(y ~ x + I(x^2))

lm(y ~ x * x) = lm(y ~ x)

lm(sec ~ f * hour * loc * pay)
$$24 \times 16 \times 3$$

e.g. Coin toss

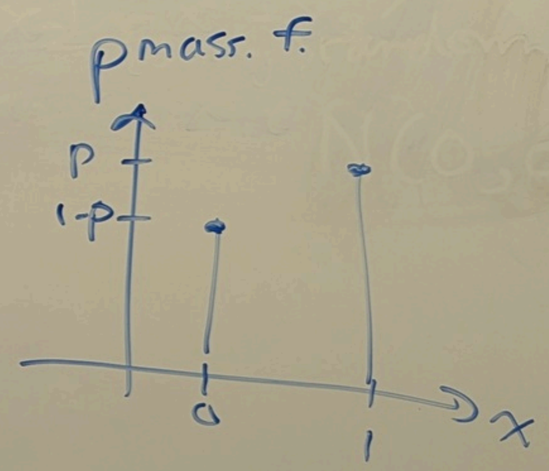
random variables $X_1, \ldots, X_n$ (1st outcome) $\sim$ independent identical dist

iid Bernoulli(p) (unknown) $0 < p < 1$

$\hat{p} = 0.45 \pm 0.00$,

$P(\text{head}) = p$

$P(\text{tail}) = 1 - p$



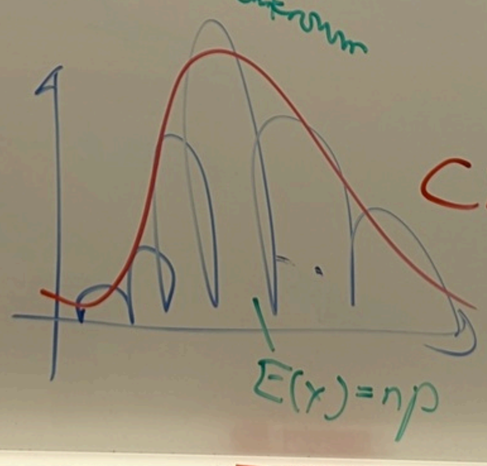
observed data $x_1, \ldots, x_n$

$Y = X_1 + \ldots + X_n \sim \text{Binomial}(n, p)$

A: # heads in n tosses; $0, 1, 2, \ldots, n$

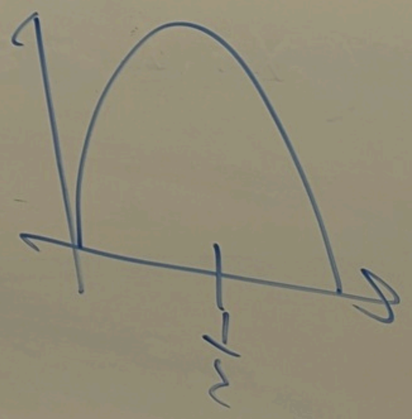
from, unknown

$E(Y) = nE(X_1) = np$



$E(X) = p$

$\text{Var}(X) = p(1-p)$

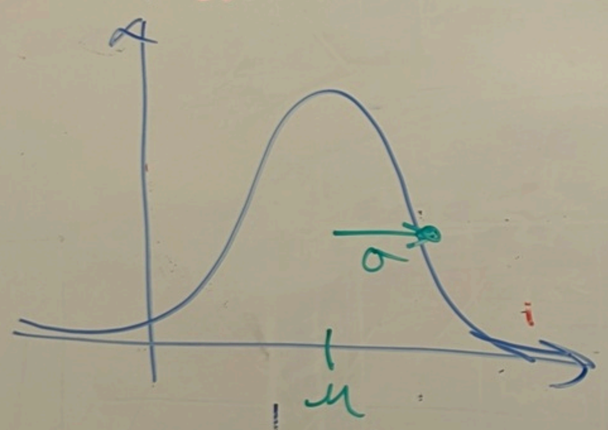


$E(X)$ $\quad$ $Var(X)$

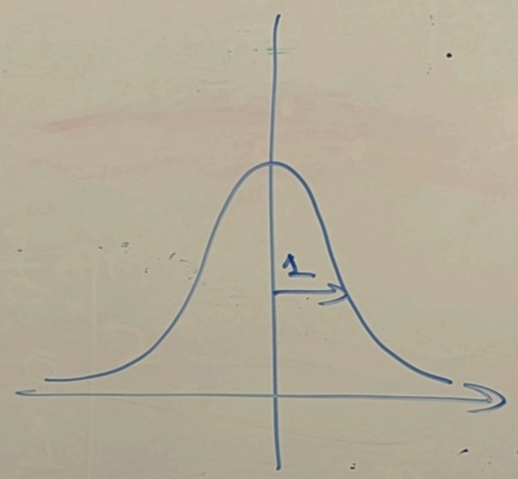
$X \sim N(\mu, \sigma^2)$

$\uparrow$ $\quad$ $\uparrow$

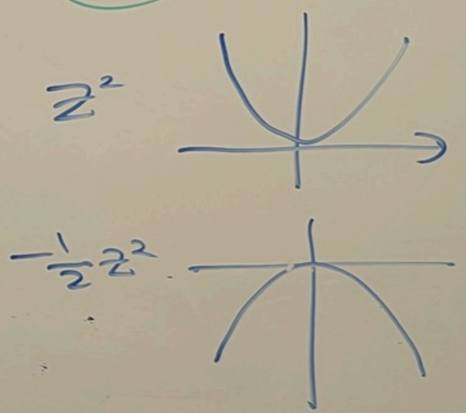
unknown



$Z \sim N(0,1)$



$$f_Z(z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{z^2}{2}}$$



Point Estimation

$X_1, \ldots, X_n \sim Bern(p)$

$\hat{p} = \bar{x} = \frac{1}{n}(x_1 + \ldots + x_n) = 0.47$

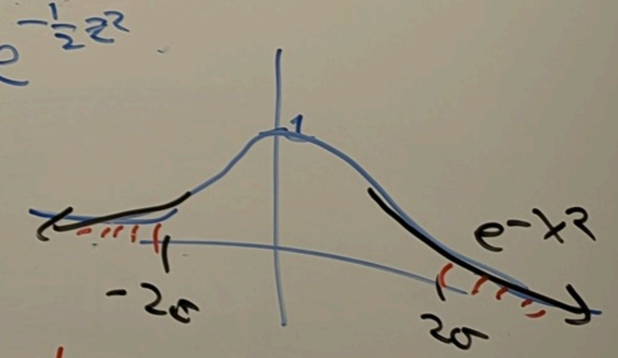
estimate

estim

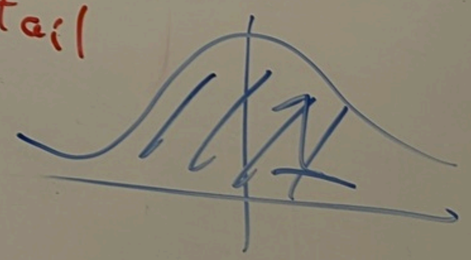
$\mu + \sigma Z \sim N(\mu, \sigma^2)$

$Var(X) = E(X - Ex)^2$

$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$



thin tail



Estimator

← procedure to compute estimate

e.g. coin toss

$$X_1, \ldots, X_n \sim \text{Bern}(p) \quad \text{(as)}$$

$$\hat{p} = \frac{X_1 + \ldots + X_n}{n}$$

(r.v.; $\hat{p}$: estimator)

$x_1, \ldots, x_n$

$$\hat{p} = \frac{0 + 1 + \ldots + 1}{n} = 0.47$$

Unbiased

$$E(\hat{p}) = p$$

(on avg; $\hat{p}$: your estimation; p: true parameter)

do this many times

works for all p

$$E(\hat{p}) = E\left(\frac{X_1 + \ldots + X_n}{n}\right)$$

$$= \frac{1}{n} E(\underbrace{X_1 + \ldots + X_n}_{nE(X_1)})$$

$$= E(X_1)$$

$$= p$$